

2025 AMC 10B Problem 11 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 11

Problem:

On Monday, 6 students went to the tutoring center at the same time, and each one was randomly assigned to one of the 6 tutors on duty. On Tuesday, the same 6 students showed up, the same 6 tutors were on duty, and the students were again randomly assigned to the tutors. What is the probability that exactly 2 students met with the same tutor both Monday and Tuesday?

Answer Choices:

A. $\frac{1}{16}$

B. $\frac{3}{16}$

C. $\frac{1}{4}$

D. $\frac{3}{8}$

E. $\frac{1}{2}$

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Solution:

Consider an assignment of students to tutors as a permutation of $[1, 2, 3, 4, 5, 6]$, where the number in the i th spot of the permutation corresponds to the student assigned to the i th tutor for $1 \leq i \leq 6$. Without loss of generality, suppose on Monday the assignment is $[1, 2, 3, 4, 5, 6]$. Then we are looking for the probability that a permutation on 6 elements has exactly 2 fixed points (spots such that the i th element is i).

There are $\binom{6}{2}$ ways to select the students to be fixed points and of the remaining 4 people, none of them can be a fixed point. This corresponds to a derangement of 4 people for a count of $4! - 4 = 9$ ways to assign the remaining 4 people.

Alternatively, if we wish to avoid using derangements, we provide another method to see that there are 9 permutations on 4 elements without fixed points.

Suppose we are arranging $[1, 2, 3, 4]$ so that i is not in the i th slot for $1 \leq i \leq 4$. Then, the first element has 3 options and there are 3 options for the spot that 1 goes to. Regardless of these two choices, there is always exactly one way to fill out the rest of the permutation, for a total of $3 \times 3 = 9$ derangements.

The final answer is then:

$$\frac{\binom{6}{2} \cdot 9}{720} = \boxed{\text{(B)} \frac{3}{16}}$$

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